

Maternal Health Risk Stratification

Statistical Analysis, Exploratory Data Analysis, and Multiclass Classification

StudyBuild | Health and Medical Data Science Project

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Abstract

This report examines the UCI Maternal Health Risk dataset using descriptive statistics, standardized group comparisons, and multinomial Logistic Regression. The project CSV contains 1,014 observations and no missing values. The initial review identified 562 exact repeated rows and two observations with HeartRate equal to 7 bpm. For the primary analysis, the two implausible HeartRate observations and all exact repeated rows were excluded, leaving 451 observations. Because patient identifiers are not available, removing repeated rows is treated as a methodological assumption rather than as proof that the rows belong to the same person. Blood Sugar showed the largest standardized difference between High Risk and the other groups, followed by systolic and diastolic blood pressure. On the primary test set, Logistic Regression achieved 0.670 accuracy. Recall was 0.565 for High Risk, 1.000 for Low Risk, and 0.048 for Mid Risk. All ten High Risk observations missed by the model were assigned to Low Risk. A separate five fold grouped sensitivity analysis retained the repeated observations while keeping identical predictor profiles within one fold. Mean accuracy in that analysis was 0.607, mean Macro F1 was 0.593, and mean High Risk recall was 0.695. The contrast between the two analyses shows that estimated performance depends on how repeated observations are handled. The fitted model is an educational baseline and is not a clinical decision tool.

Keywords: maternal health; risk stratification; blood sugar; blood pressure; logistic regression; recall; error analysis; grouped validation.

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1. Introduction

Maternal health risk classification deserves careful evaluation because physiological measurements vary across individuals and because the practical meaning of an error depends on the class involved. A High Risk observation assigned to Low Risk is more concerning than many other classification errors. For that reason, this project reports class specific performance and examines misclassified observations rather than relying on accuracy alone.

The main question is whether the six recorded physiological measurements differ across Low Risk, Mid Risk, and High Risk groups, and how well a simple multinomial Logistic Regression model can distinguish those categories.

The analysis starts with a review of the source data, including missing values, repeated rows, class balance, and unusual measurements. Descriptive summaries and standardized mean differences are then used to compare the risk groups. A Logistic Regression model is fitted to a stratified training sample, evaluated on a held out test set, and followed by error analysis and a grouped sensitivity analysis for repeated observations.

2. Data and Study Design

2.1 Data source and variables

The project uses the UCI Maternal Health Risk dataset, Dataset 863. The StudyBuild project brief reports 1,013 instances in the UCI version, while the CSV used here contains 1,014 rows. The analysis uses the supplied project file and records this difference as a source version issue.

Variable	Unit / Type	Description
Age	years	Maternal age
SystolicBP	mmHg	Systolic blood pressure
DiastolicBP	mmHg	Diastolic blood pressure
BS	mmol/L	Blood sugar
BodyTemp	°F	Body temperature
HeartRate	bpm	Heart rate
RiskLevel	categorical	Low Risk, Mid Risk, or High Risk

2.2 Data review and analytical sample

The raw file contains 1,014 observations and seven variables, with no missing values. The main data quality findings and the decisions used for the primary analysis are summarized below.

Check	Finding	Decision
Missing values	0	No imputation was needed.
Exact repeated rows	562	Removed in the primary analysis to reduce the chance that identical rows are represented in both training and test data. Patient identifiers are unavailable, so this remains a methodological assumption.
HeartRate = 7 bpm	2 rows	Excluded because the values are physiologically implausible for this dataset.
Very young ages	Minimum age = 10	Kept because the available data do not provide enough evidence to classify them as recording errors.
Conflicting labels	35 identical predictor profiles	Kept. The same recorded predictors can be linked to different RiskLevel labels, which may reflect missing clinical information or label uncertainty.
Final modeling sample	451 observations	Used for descriptive analysis, SMD calculations, the primary training and test split, and Logistic Regression.

Removing exact repeated rows changed the class distribution. Low Risk increased from 40.0% of the raw data to 51.7% of the primary modeling sample, while Mid Risk decreased from 33.1% to 23.5%. The repeated rows were therefore not distributed evenly across the labels. This change is important when the primary results are interpreted.

3. Exploratory Data Analysis

3.1 Risk level distribution

Risk Level	Count	Percent
Low Risk	233	51.7%
Mid Risk	106	23.5%
High Risk	112	24.8%

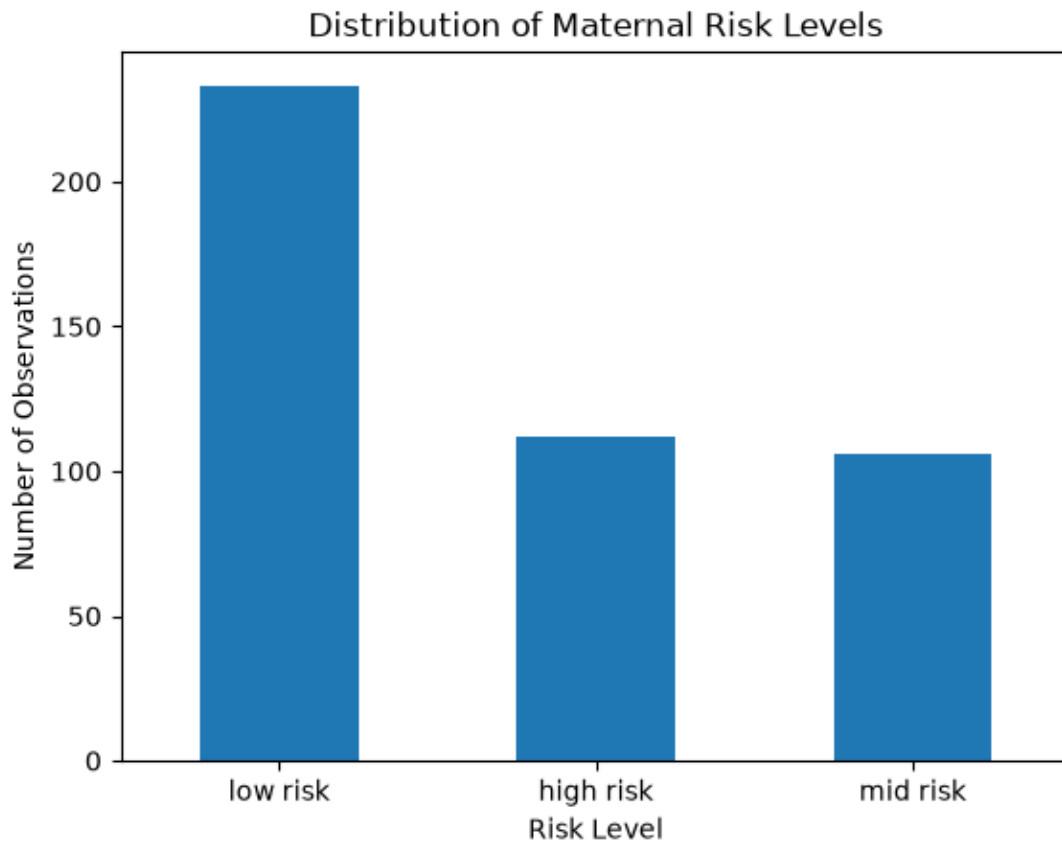


Figure 1. Distribution of Low Risk, Mid Risk, and High Risk observations in the primary modeling sample.

The three classes are moderately imbalanced. Low Risk accounts for about half of the primary modeling sample, while High Risk and Mid Risk are similar in size. Class specific metrics are needed because the frequency and the difficulty of the three categories differ.

3.2 Clinical patterns across risk groups

Risk	Age	SBP	DBP	BS	BodyTemp	HeartRate
Low Risk	27.37	105.37	72.72	7.20	98.36	73.04
Mid Risk	28.54	112.41	74.89	7.89	98.86	73.90
High Risk	33.73	119.49	81.54	11.17	99.23	76.48

The High Risk group has higher mean values for all six recorded measurements in the primary sample. The largest descriptive difference is seen in Blood Sugar, followed by systolic and diastolic blood pressure. The group distributions still overlap, so none of these measurements separates the three labels by itself.

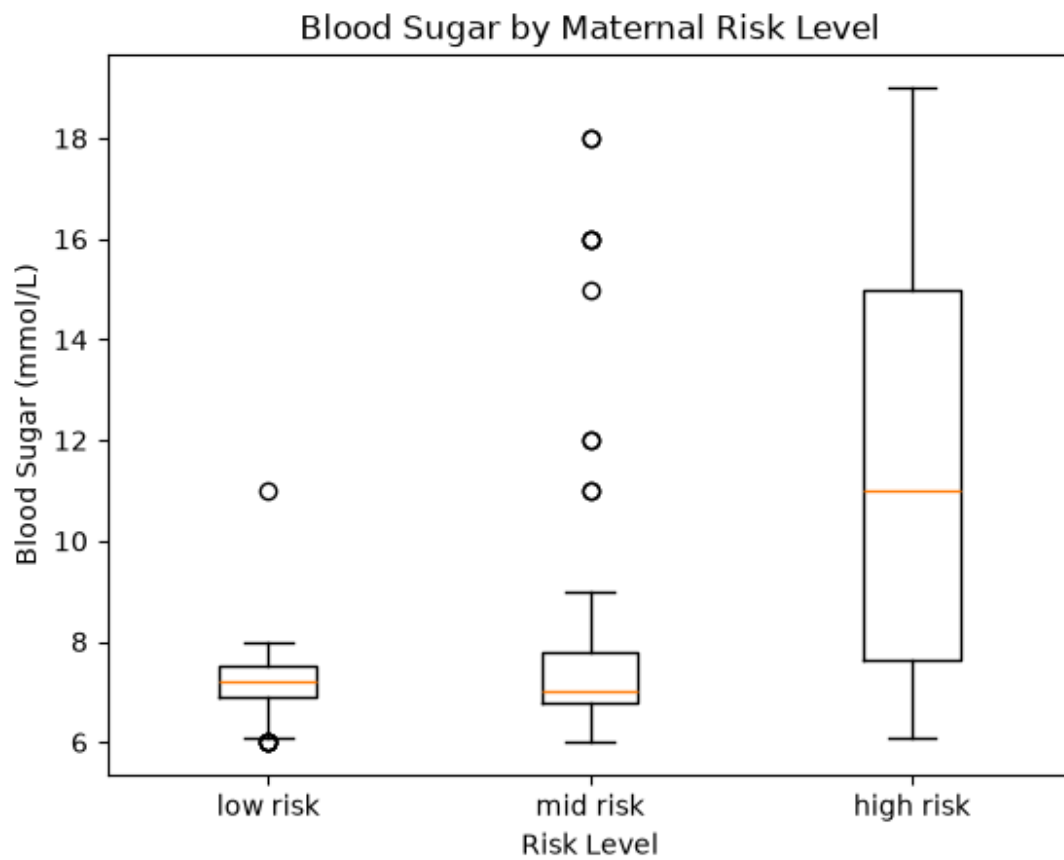


Figure 2. Blood Sugar distribution by maternal risk group.

Blood Sugar shows the clearest visual difference among the three groups. High Risk contains a larger share of elevated BS values, although lower values also occur in that group.

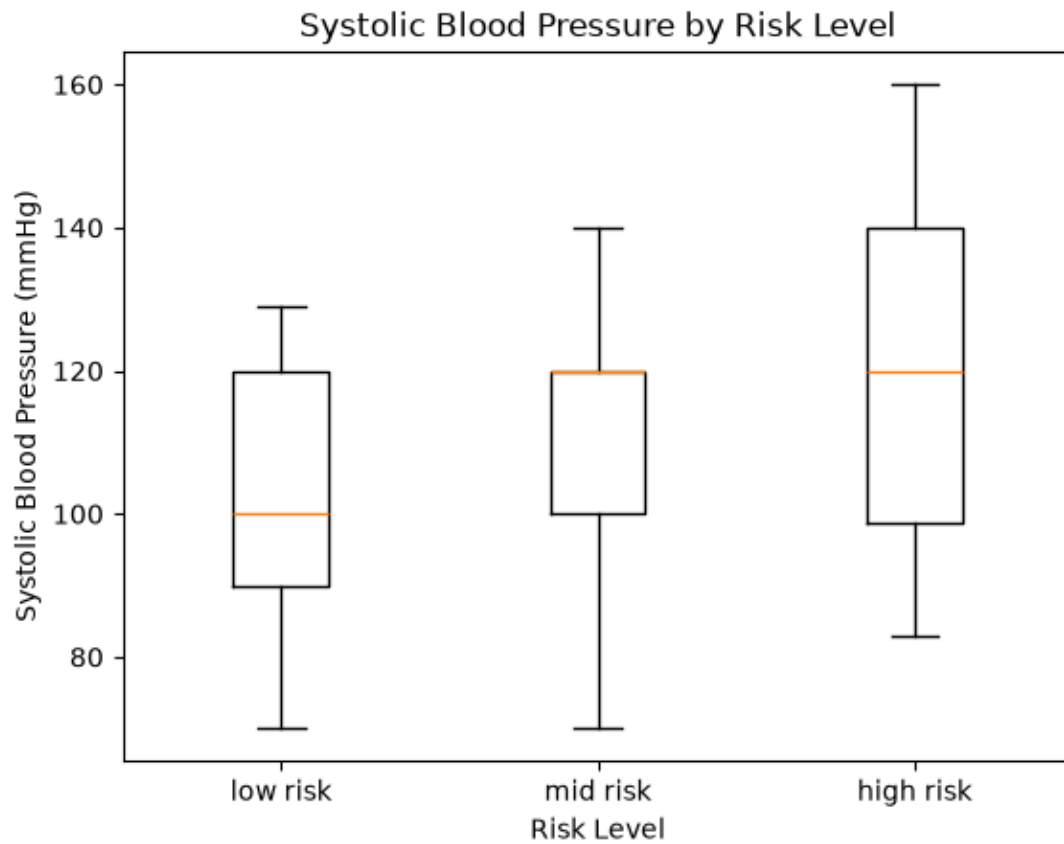


Figure 3. Systolic blood pressure distribution by maternal risk group.

Systolic blood pressure is generally higher in High Risk, but the distributions overlap considerably. A single pressure threshold would not reproduce the three RiskLevel categories in this dataset.

3.3 Standardized differences for High Risk

The predictors use different units, so raw mean differences are not directly comparable. High Risk was therefore compared with the combined Low Risk and Mid Risk groups using absolute Standardized Mean Differences, or SMDs. Each SMD expresses the group difference relative to pooled variability.

Clinical Feature	Absolute SMD: High Risk vs Other
Blood Sugar	1.613
Systolic BP	0.695
Diastolic BP	0.611
Body Temperature	0.515
Age	0.443
Heart Rate	0.428

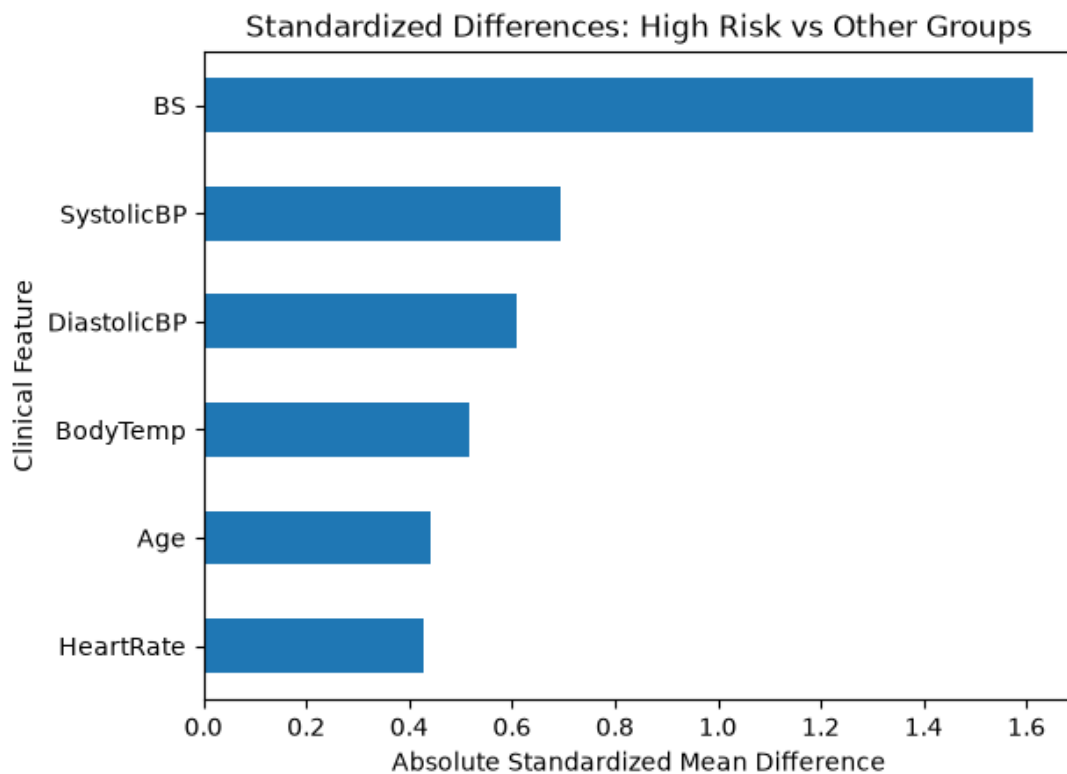


Figure 4. Absolute Standardized Mean Differences for High Risk compared with the other two groups.

Blood Sugar has the largest standardized difference, followed by systolic and diastolic blood pressure. These values describe separation in this dataset and do not establish causal clinical effects.

4. Statistical Methods and Classification Model

4.1 Training and test design

The 451 observation primary dataset was divided into 80 percent training data and 20 percent test data with random state 42. Stratification kept the three RiskLevel proportions similar in both subsets. The split produced 360 training observations and 91 test observations.

4.2 Logistic Regression pipeline

Multinomial Logistic Regression was selected as a simple and interpretable baseline for the three category outcome. StandardScaler and LogisticRegression were fitted in one Pipeline. Scaling parameters were therefore estimated from the training data and then applied to the test data, so preprocessing did not use information from the test set.

```
Pipeline([
    ("scaler", StandardScaler()),
```

```
("logreg", LogisticRegression(max_iter=1000))
])
```

4.3 Evaluation strategy

The primary model was evaluated with accuracy, precision, recall, F1 score, and a confusion matrix. High Risk recall was examined closely because it measures the proportion of true High Risk observations identified by the model. A separate sensitivity analysis retained repeated observations and used five fold StratifiedGroupKFold validation, with identical predictor profiles kept within the same fold. This second analysis was used to assess how strongly the performance estimates depend on the treatment of repeated rows.

5. Classification Results

5.1 Primary test set performance

Class	Precision	Recall	F1	Support
High Risk	0.812	0.565	0.667	23
Low Risk	0.635	1.000	0.777	47
Mid Risk	1.000	0.048	0.091	21

Accuracy on the primary test set was 0.670 and Macro F1 was 0.511. Performance differed sharply by class. Recall was 1.000 for Low Risk, 0.565 for High Risk, and 0.048 for Mid Risk.

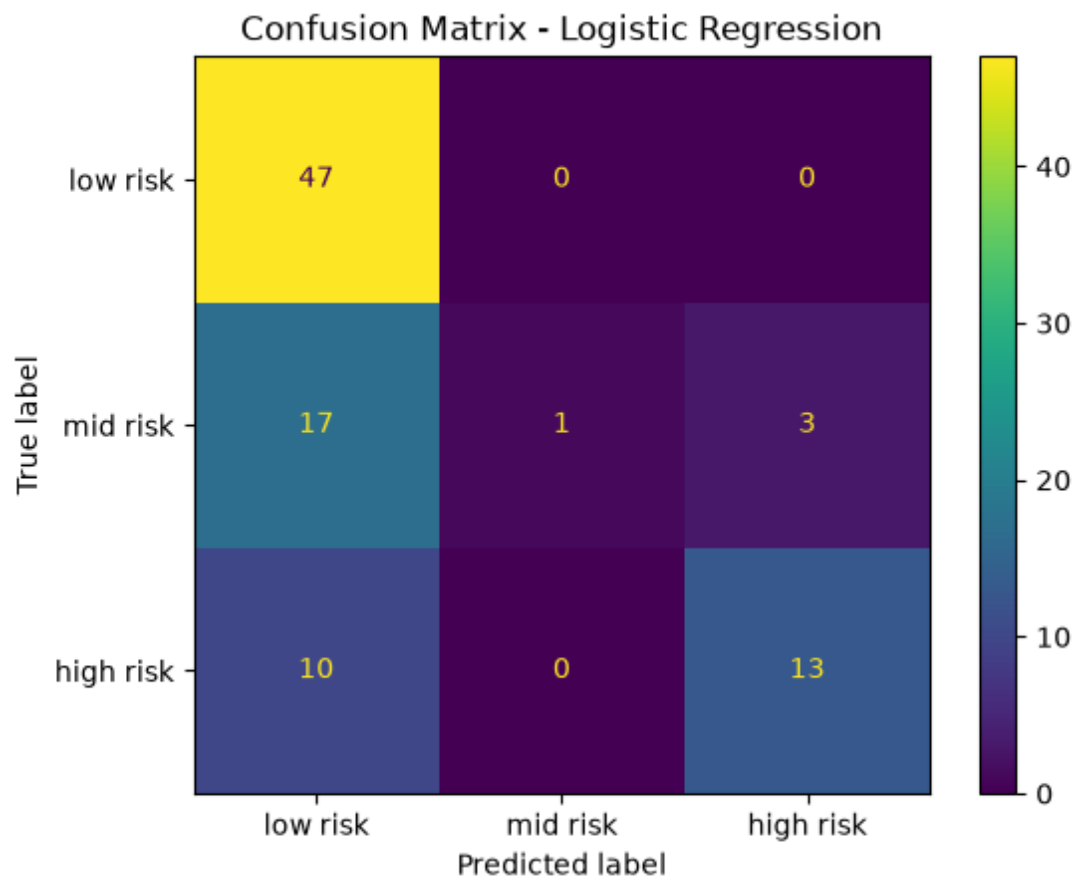


Figure 5. Confusion matrix for the multinomial Logistic Regression baseline.

Among 23 true High Risk observations, 13 were identified correctly and 10 were missed. The High Risk miss rate in this test set was therefore about 43.5%. Mid Risk was more difficult: only 1 of 21 observations was classified correctly.

5.2 Error analysis

Error Type	Count
Mid Risk to Low Risk	17
High Risk to Low Risk	10
Mid Risk to High Risk	3

The model often assigned uncertain observations to Low Risk. All ten missed High Risk observations were classified as Low Risk. This pattern is important because it shows a specific weakness that is not visible from the overall accuracy value.

Feature	Missed High Risk	Correctly Identified High Risk
Age	33.30	36.77
SystolicBP	117.00	124.23
DiastolicBP	84.50	84.62
BS	7.35	12.81
BodyTemp	98.50	99.08
HeartRate	75.50	80.92

The clearest difference between missed and correctly identified High Risk observations was Blood Sugar. Mean BS was 7.35 mmol/L among the missed cases and 12.81 mmol/L among those identified correctly. In this test set, High Risk observations without markedly elevated BS were harder for the model to recognize.

6. Model Coefficients

The predictors were standardized before Logistic Regression was fitted, so their coefficient magnitudes can be compared more meaningfully. The coefficients describe relationships within the fitted prediction model and should not be interpreted as causal clinical effects.

Feature	High Risk Standardized Coefficient
Blood Sugar	+1.176
Body Temperature	+0.524
Systolic BP	+0.345
Age	-0.214
Diastolic BP	+0.178
Heart Rate	+0.166

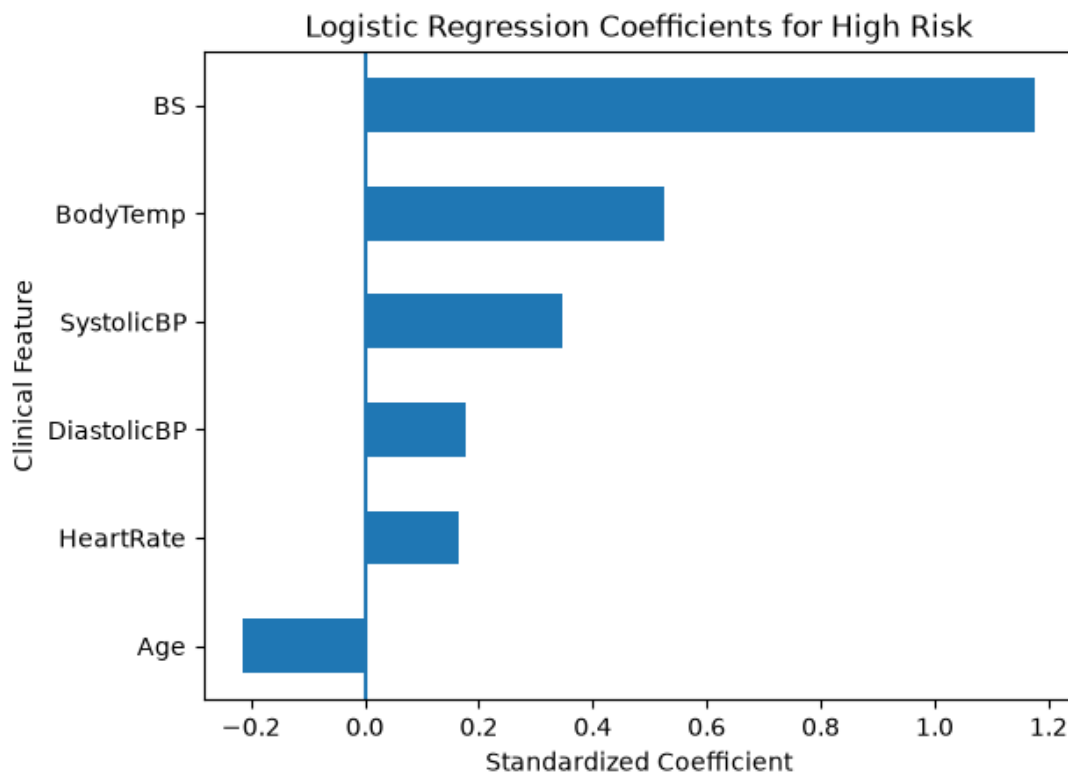


Figure 6. Standardized Logistic Regression coefficients for High Risk.

Blood Sugar has the largest positive coefficient for High Risk, followed by Body Temperature and Systolic BP. The negative coefficient for Age does not mean that older age is protective. In multinomial Logistic Regression, each coefficient is estimated in the presence of the other predictors, and its sign can be affected by overlap and correlation among variables.

7. Discussion

The descriptive analysis and the fitted model point to the same broad pattern: Blood Sugar provides the largest separation for High Risk, while blood pressure contributes additional information. At the same time, the class results show that the six predictors do not separate the three risk categories cleanly.

The primary holdout result is not the only plausible estimate of performance. When repeated observations were retained and identical predictor profiles were kept within a single fold, five fold grouped validation produced mean accuracy 0.607 ± 0.029 , mean Macro F1 0.593 ± 0.032 , mean High Risk recall 0.695 ± 0.057 , and mean Mid Risk recall 0.330 ± 0.139 . The maximum overlap of identical predictor profiles between training and test data was zero in every fold.

The primary and grouped results use different samples and different validation procedures, so they should not be treated as a direct competition. Their main contribution is to show that the reported performance depends on

how repeated observations are handled. This matters because the source data do not contain patient identifiers and because 35 identical predictor profiles are linked to more than one RiskLevel label.

8. Limitations

- The primary modeling sample contains 451 observations after the selected preprocessing steps.
- Only six predictors are available. Other clinical information that could matter for maternal risk is not included in this dataset.
- Patient identifiers are unavailable, so identical rows cannot be confirmed as repeated records from the same person. Removing them is therefore a methodological assumption, and it changes the class proportions.
- Some identical predictor profiles have different RiskLevel labels. This may reflect missing clinical information, uncertainty in the labels, or limitations in the original labeling process.
- The project CSV contains 1,014 rows, while the UCI description reports 1,013 instances. The available project materials do not explain this difference between versions.
- The primary model is evaluated on one stratified training and test split. The five fold grouped analysis gives a useful sensitivity check, but neither analysis provides external validation.
- RiskLevel is an existing category in the source dataset rather than a prospectively observed clinical outcome.
- The observed associations and fitted coefficients should not be interpreted as causal clinical relationships.

9. Recommendations and Future Work

- Obtain patient identifiers or information about repeated measurements if a future version of the dataset becomes available. This would allow repeated records to be handled on a stronger empirical basis.
- Validate the model on an independent dataset before drawing conclusions about performance in another population.
- Include additional clinical variables if they are available, especially information that may help distinguish Mid Risk from Low Risk.
- Investigate how RiskLevel was assigned in the source data and whether the six predictors were part of the original labeling rule.
- Compare the Logistic Regression baseline with one simple nonlinear model, such as a shallow Decision Tree, while keeping class specific evaluation and error analysis unchanged.
- For future screening work, report uncertainty around High Risk recall and consider whether different types of classification error should carry different practical costs.

10. Conclusion

Blood Sugar shows the largest standardized difference for High Risk in this dataset, followed by the blood pressure variables. The primary Logistic Regression model achieved 67.0% accuracy, but its class performance was uneven: High Risk recall was 56.5% and Mid Risk recall was 4.8%. Error analysis showed that all ten missed High Risk observations were assigned to Low Risk.

The grouped sensitivity analysis adds an important qualification. Keeping repeated observations while preventing identical predictor profiles from crossing folds changed the estimated performance, including higher average recall for High Risk and Mid Risk. The main lesson from the project is therefore methodological: model results should be read together with the data preparation choices, the class specific errors, and the limitations of the source data. The fitted model remains an educational baseline and should not be used for clinical decisions.

References

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